Simultaneous Screening and Detection of RFI From Massive SAR Images: A Case Study on European Sentinel-1

Ning Li, *Member, IEEE*, Hengrui Zhang, Zongsen Lv, *Graduate Student Member, IEEE*, Lin Min, and Zhengwei Guo, *Member, IEEE*

***Abstract*— Currently, the spaceborne synthetic aperture radar (SAR) system transmits a great deal of data to the ground processing station and generates massive images daily, only a tiny fraction of which contains radio frequency interference (RFI). However, most of the existing RFI detection methods are based on the prior conditions in which the known image contains interference. In fact, it is difficult to learn whether SAR images contain RFIs without prescreening, so it is of great significance to the rapid and real-time screening and detection of RFI in SAR images. This article proposes a method to screen and detect RFI from massive SAR images simultaneously. 1) We construct an approximate RFI-free background image by using the preprocessed time-series SAR images acquired in the past. 2) We generate difference images based on the image change detection method and analyze them by using an adaptive threshold, then calculate the entropy of all difference images to complete the preliminary screening of the RFI-containing). According to the preliminary results, we remove the RFI-containing parts; after reconstructing the background, we repeat step 2 with the images to be detected and obtain the final screening and detection results. Massive experimental results based on Sentinel-1 images validate the performance of the proposed method.**

***Index Terms*— Image change detection, interference detection, radio frequency interference (RFI), synthetic aperture radar (SAR).**

I. Introduction

A. Background

SYNTHETIC aperture radar (SAR), an active microwave remote sensing device with all-day and all-weather characteristics, can obtain high-resolution remote sensing images in terms of range and azimuth by transmitting high-bandwidth signals and combining with the synthetic aperture formed by moving along the track [1]–[6]. Time-series SAR images have rich spatial and temporal information and are characteristically massive, high-dimensional, and unstructured. This property makes it unique and advantageous in land planning [7], urban expansion [8], natural disaster assessment [9], agricultural surveys [10], and ecological environment monitoring [11], so related research and applications have received extensive attention.

However, with the development of global radio communications, the electromagnetic environment has become more complex. SAR is susceptible to various convoluted electromagnetic interferences in the operating frequency band as an open broadband radar system. In general, the signals emitted by other radiation sources in the same frequency band are called radio frequency interference (RFI) to SAR [12]–[17]. The existence of RFI will seriously restrict the high-resolution imaging effect and further affect the subsequent applications of SAR data, such as crop monitoring and natural disaster assessment [18]. Uncovering how to survive in this complex electromagnetic environment while maintaining the excellent performance of SAR as much as possible has become a vital research trend in the SAR field [19]–[22].

In recent years, many cases of RFI have been found in spaceborne SAR systems. Fig. 1 shows the RFI probability map of Sentinel-1 spaceborne SAR systems operating in the C-band [23]. In terms of probability, only a tiny fraction of spaceborne SAR products contain RFI. However, in the field of SAR interference research, most studies are based on the prior knowledge that the data contain RFI, and few pieces of literature mention how to screen the RFI-containing part from a large amount of data quickly [24]. In addition, the existing RFI detection methods are mainly applied to SAR level-0 products, which are difficult to apply to spaceborne SAR that generally only provide level-1 and above products to public access. Therefore, it is of great significance to research methods to screen and detect RFI from massive SAR images.

B. Previous Work

Effectively detecting RFI in SAR data has always been an important research topic in the SAR field. Since the 1990s, various SAR interference detection methods have been proposed [25], and they can be placed into two categories: RFI detection methods based on level-0 products and those based on level-1 products.

1) RFI Detection Methods Based on Level-0 Products: The RFI detection methods based on the level-0 products use the features of RFI in the time domain, frequency

Manuscript received 24 April 2022; revised 10 June 2022; accepted 13 July 2022. Date of publication 18 July 2022; date of current version 26 July 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61871175; in part by the Foundation of the Key Laboratory of Radar Imaging and Microwave Photonics, Ministry of Education, under Grant RIMP2020003; and in part by the Graduate Education Innovation and Quality Improvement Program of Henan University under Grant SYL20060144. *(Corresponding author: Ning Li.)*

The authors are with the College of Computer and Information Engineering, the Henan Key Laboratory of Big Data Analysis and Processing, and the Henan Engineering Research Center of Intelligent Technology and Application, Henan University, Kaifeng 475004, China (e-mail: lining_nuaa@163.com; hedalining@henu.edu.cn; zhr_henu@henu.edu.cn; lvzongsen@henu.edu.cn; mlin@henu.edu.cn; gzw@henu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3191815

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

domain, or other transform domain to complete the detection work. The time-domain interference detection methods can effectively detect sine wave-like interference signals. Based on the property in which a strong interference signal will cause some statistical characteristics of the entire SAR echo to be abnormal, the moment kurtosis coefficient is calculated to detect whether there is RFI in the echo [26]. However, this statistical feature-based interference detection method cannot give the spectrum distribution of the interference. Recently, a study in [27] proposed a time-domain detector (TDD) for detecting and locating pulse RFI (PRFI), which is based on the characteristics of PRFI with strong energy and short duration in the time domain. The adjacent pulse change detection technique is used to detect the position of interference. However, this method has poor performance when the RFI is weak.

Transform-domain interference detection methods use the different features of background noise, received signals, and interference signals in the transform domains. They primarily include frequency domain, wavelet transform, discrete cosine transform, lapped transform, and other interference detection methods. The transform domain has many advantages, e.g., the complex filtering process in the time domain can be achieved by multiplication in the transform domain. The lowpass and bandpass filters can be easily applied in the transform, which cannot be performed in the time domain [28]–[30]. In 1998, for powerful interference signals, a transform domain detection method based on the spectrum in range-frequency domain was used in [31]. The short-time Fourier transform (STFT) is performed in the azimuth-time and range-frequency domain to obtain the average amplitude spectrum for detecting interference. However, when the target backscatters were strong, this method may not work. To overcome the disadvantage of the reduced frequency resolution in the range spectrum method, in 2003, the average range spectrum method, which averages the range spectrum of a group of adjacent pulses, was proposed in [32]. This method employs the strong coherence of RFI in the range spectrum to detect interference. In 2007, eigenvalues were used as a tool for interference detection based on the uncorrelated properties of echo signals and RFI in [33]. By transforming the echo space into a signal subspace and an RFI subspace using eigenvalue decomposition (EVD), the presence of interference can be successfully detected only by judging whether there is a subspace composed of interference signals. In 2011, based on the average range spectrum and the kurtosis coefficient detection method, combined with the azimuth distribution characteristics of the spaceborne SAR interference signal, an improved average range spectrum-subband moment kurtosis method was proposed in [34]. This method combines the characteristic parameters of the interference signal and adopts the secondary detection method, which reduces the difficulty of setting the threshold of the average range spectrum method and improves the detection probability simultaneously. In 2017, the local autocorrelation-based detection method was adopted in the range-time and azimuth-frequency domains in [35], which can effectively detect intermittent transmission broadband RFI. In 2018, a method to detect and geolocate RFIs using digital beamforming (DBF) was proposed in [36], which was implemented by using NASA's EcoSAR measurements. Among them, the detection work of RFI is performed by a temporal bandpass filter, and the DBF is used to estimate the angle of arrival and thus geolocate the interference. In 2022, a time-frequency detector (TFD), based on a two-step process combining EVD with STFT was proposed in [37]. In contrast to the TDD, the proposed detector can detect and locate PRFIs submerged in the SAR echo of strong targets by accumulating the energy of PRFIs in the time-frequency domain.

It is worth noting that most of the methods mentioned above based on raw data involve a complex spectrum analysis process and are often based on the premise that the known data contain interference to detect the pulse position where interference occurs. For spaceborne SAR, which generally only provides level-1 and above products to public access, these methods cannot be directly applied to the screening and detection of RFI images.

2) RFI Detection Methods Based on Level-1 and Above Products: With the increasing availability of spaceborne SAR data [38], [39], orbiting satellites with SAR systems transmit a large amount of data to ground processing stations and generate massive images every day. By contrast, manual screening for RFI images and contaminated areas is time-consuming and challenging to guarantee accuracy. At present, some scholars have noticed the importance of image-based RFI detection technology and have proposed some detection methods.

In 2020, based on image processing methods, interference information was initially extracted from Sentinel-1 quick-look images with RFI by using a convolutional neural network as a reference classifier in [40]. In 2021, based on the deep convolutional neural network to extract image features, a set of Sentinel-1 processing procedures for quick-look images with RFI was proposed in [41], which is used to identify different RFI levels in SAR images. However, these two methods are based on the quick-look image for RFI detection. Due to the low resolution of the image set, it is difficult to guarantee the accuracy of RFI detection. In addition, these deep learning methods require numerous samples for model training, which are inefficient for interference detection in a large number of single look complex (SLC) or ground range detected (GRD) products.

In 2021, the interference features in dual-polarization SAR images were fully put to use, and based on the ascending and descending orbit images, a set of complete and operable processes of RFI detection and localization in Sentinel-1 images was proposed in [42]. This processes can be used to detect and locate diamond-shaped ground interference sources with a range of approximately 88.76 km^2. However, based on the prior condition that the known data contain interference, this method is used to perform interference detection on a single image and locates the possible location of the ground interference source.

In 2021, a joint low-rank and sparse model for RFI extraction and mitigation based on a coupled analysis of time-series images was proposed in [43]. This method exploits the spatiotemporal correlation of land cover in SAR images and the time-varying nature of RFI artifacts, enabling the extraction of RFI signatures from Sentinel-1 quick-look images and mitigating the effects of RFI. However, it also detects and mitigates

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

RFI in time series based on the premise that the data contain RFI. Since the detection process involves matrix singular value decomposition (SVD) and the setting of hyperparameters, it is not suitable for the prescreening of RFI in massive SAR images.

In February 2022, to cope with the impact of interference on SAR image quality, the European Space Agency (ESA) upgraded the Sentinel-1 satellite ground processor and added the interference detection module [44], which explicitly includes interference prescreening and detection functions. Interference prescreening is used to screen out the data containing interference from the massive SAR data quickly. However, this step requires the SAR system to receive additional "noise signals," which requires targeted system design. Interference detection is based on distribution statistics, which are used to locate the position of interference in echo data. This operation is simple to implement, but the application accuracy is low due to the variety of interference types. Therefore, it is of great application value to develop a method that can quickly screen and detect RFIs from massive SAR images.

C. Solution and Contributions of This Article

To provide a suitable approach to solving the above problems, this article employs Sentinel-1 GRD products as an example and proposes a novel idea for screening and detecting RFI from SAR images. First, the radiometric calibration and image registration preprocessing are performed on the time-series images, and the background of the research area is constructed. Second, based on the image change detection method, the difference images are generated and analyzed by the adaptive threshold, then calculate the entropy of all difference images to complete the preliminary screening of the RFI-containing images. Lastly, according to the primary screening results, the RFI-containing parts are removed. After the background is reconstructed, the above step is repeated with the images need to be detected, and the final screening and detection results can be obtained. This method can quickly screen RFI-containing parts and detect RFI signatures from massive SAR images. The performance and potential of this method are verified by a large number of experimental results based on Sentinel-1 Level-2 products.

The main contributions of this article are summarized as follows.

1) The screening of SAR images containing RFI based on Level-2 products of spaceborne SAR is implemented in this article, which only needs to use the existing images to establish an RFI-free background to complete the subsequent screening of massive SAR images. To the best of our knowledge, this idea is the first to be applied to the field of SAR RFI detection.
2) A novel detector based on an image change detection operator, which has the capability to detect RFI signatures in batches from massive SAR images quickly, is proposed. Compared with traditional RFI detection methods, the proposal independent of the phase and other information in the level-0 products, which can simultaneously complete the screening of RFI-contaminated images and the detection of RFI artifacts.
3) Experiments on the RFI screening and detection of Sentinel-1 images in Beijing, Kaifeng, Hangzhou, China, and parts of the Arabian Peninsula are presented and discussed. The results show that the method can be used to construct a RFI-free background database to screen whether the images in the study area contain interference batch-by-batch and detect the RFI signature. It provides a robust and efficient approach for evaluating the quality of spaceborne SAR data and detecting RFI.

D. Organization of This Article

The rest of this article is organized as follows. Section II introduces the geometric and signal models of RFI and analyzes the characteristics of RFI in Sentinel-1 images. Section III briefly introduces the basic theory on the related work of the proposed method. Section IV details the workflow of the proposed RFI screening and detection method. Section V provides a detailed introduction to the datasets used in the experiments. Section VI presents the experimental results and analysis, followed by a discussion of further applications of the method. Lastly, Section VII concludes this article.

II. Models and Characteristics of RFI

A. Geometric Model of RFI

The main sources of RFI in SAR images are many different radiation sources operating in the same frequency band as the SAR system. In recent years, many RFI cases have been observed in Sentinel-1 data products, including ground-based sources such as ground-based fixed base stations, mobile base stations, or ground-based radars, and space-based sources such as airborne radars and co-frequency satellites. Fig. 2 shows the spatial relationship between the interference source and the irradiated area of the spaceborne SAR beam. For ground-based sources, the interfering signal is received in the form of direct waves by the main or sidelobes of the SAR. For space-based sources, the interfering signal is received by the SAR in the form of direct or scattered waves.

B. Signal Model of RFI

While there are many different forms of RFIs, in general, an RFI signal is considered to be a linear combination of multiple single frequencies caused by a radiating source. According to the bandwidth, it can be divided into two categories: narrow-band interference (NBI) and wide-band interference (WBI). As mentioned in [24], NBI can be regarded as the superposition of a series of sinusoidal signals, and its model can be expressed as

$$\mathrm{RFI}_{\mathrm{NB}}(\tau, \eta) = \sum_{n=1}^{N} A_n(\eta) \exp(2j\pi f_n \tau + \varphi_n) \tag{1}$$

where N represents the number of RFI signals. $A_n(\eta)$, f_n, and φ_n represent the amplitude, frequency, and phase of the nth RFI signal, respectively. NBI generally does not have complex frequency modulation and often appears in the form of bright lines in the image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

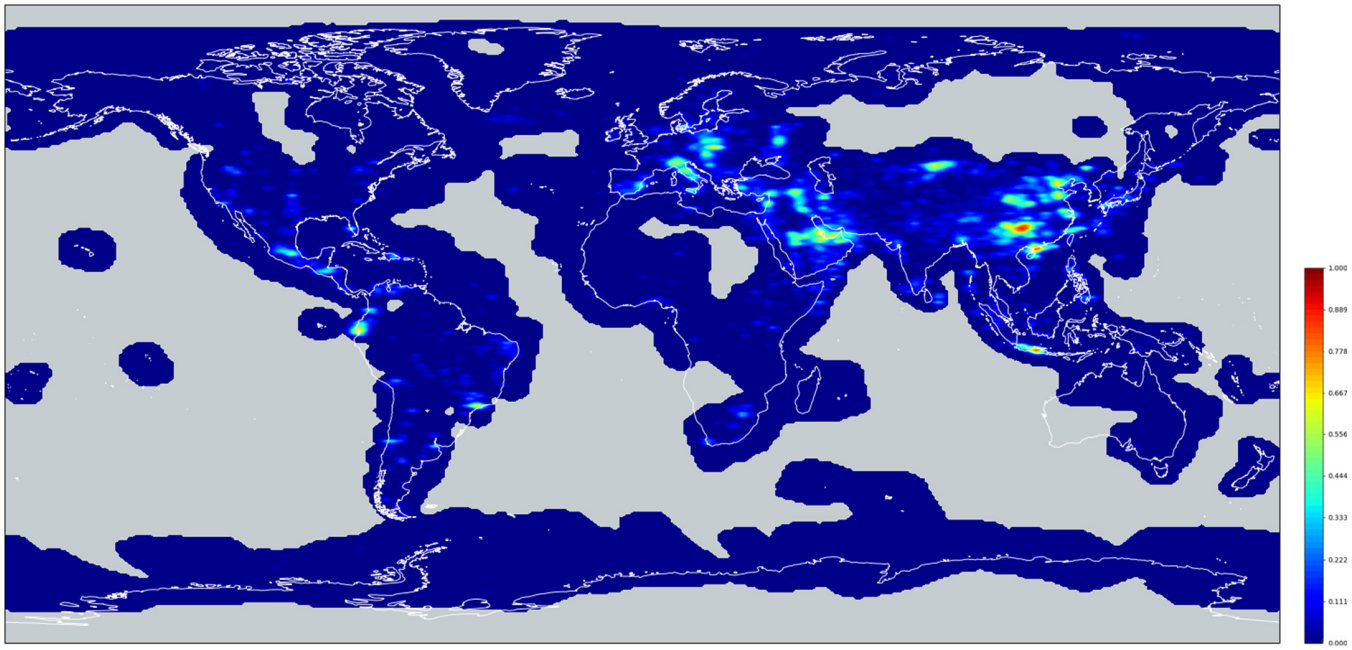


Fig. 1. Probability distribution map of the RFI of Sentinel-1 spaceborne SAR system in the global area [23].

By contrast, the frequency modulation of WBI is more complex and has a large bandwidth, which appears as blur or blurry stripes in the image. Without loss of generality, WBI can be modeled as linear frequency modulation (LFM) and sinusoidal frequency modulation (SFM) [27]. Its signal model can be expressed as

$$\mathrm{RFI}_{\mathrm{WB}}(\tau,\eta)=\sum_{n=1}^{N}A_n(\eta)\exp(j\phi) \tag{2}$$

where the phase term ϕ is divided into two types of phase modulation terms ϕ_{CM} and ϕ_{SM}, LFM and SFM, respectively, expressed as

$$\phi_{\mathrm{CM}}=2\pi f_n\tau+\pi K_n\tau^2 \tag{3}$$

$$\phi_{\mathrm{SM}}=\beta_n\sin(2\pi f_n\tau+\varphi_n) \tag{4}$$

where K represents the chirp rate of the nth RFI signal, and β represents the modulation factor.

Since the RFI signal is independent of the SAR echo signals, the SAR echo data with interference can be expressed as

$$R(\tau,\eta)=S(\tau,\eta)+N(\tau,\eta)+\mathrm{RFI}(\tau,\eta) \tag{5}$$

where $S(\tau,\eta)$, $N(\tau,\eta)$, and $\mathrm{RFI}(\tau,\eta)$ represent the useful echo signal, system noise, and RFI signal, respectively. τ and η represent the fast time in the range direction and the slow time in the azimuth direction, respectively.

C. Characteristics of RFI in SAR Images

The main detrimental effect of RFI on SAR is that it can introduce radiation artifacts into the focused SAR image, thereby reducing image quality. The characteristics of RFI in SAR images can be summarized in the following three points.

First, RFI causes energy suppression in SAR images. In general, the power of RFI is stronger than radar echo, clutter, and system noise. Even though SAR can obtain higher compression gain through 2-D matched filtering, RFI still exhibits different degrees of energy suppression in SAR images. The suppression degree is related to factors such as the type of interference source and the interference path.

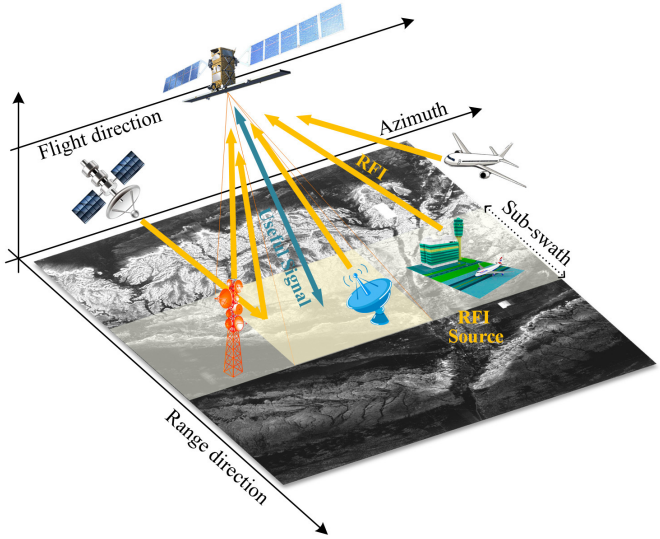


Fig. 2. Common sources and geometric model of RFI.

Second, RFI produces artifacts in SAR images. In SAR images, the RFI after imaging processing is usually present in the form of artifacts, which are manifested as various linear image features, such as bright stripes with curvature, fog or raindrops, bright stripes, etc., [45]; some artifacts also have a specific periodicity.

Third, RFI randomness in SAR images. Since the working time of the interference source is not fixed, whether the SAR images obtained at different times in the same area contain RFI and the intensity of RFI are random. In addition, due to the pulse repetition frequency of RFI being different from that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

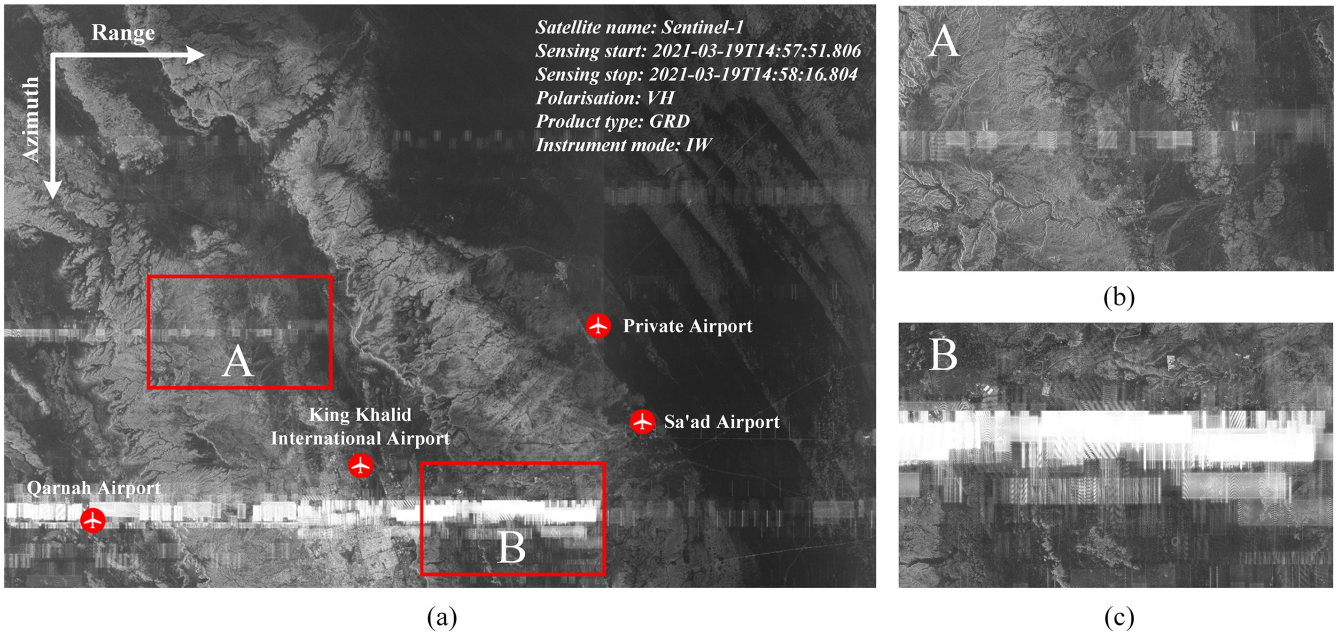


Fig. 3. Example case of the SAR image contaminated by RFI. (a) SAR image contaminated with RFI. (b) Region of interest A, artifact produced by RFI. (c) Region of interest B, energy suppression properties of RFI.

of SAR, only when the RFI appears in the SAR receiving time will it affect the image, which means that the position of RFI in the SAR image is random.

To characterize RFI in images, two example case studies, i.e., spaceborne SAR images contaminated with RFI and a time-series dataset collected by Sentinel-1, are shown in Figs. 3 and 4.

1) Example Case on the Energy Suppression and Artifact of RFI in SAR Images: Fig. 3 shows an example of a Sentinel-1 image contaminated by RFI. As shown in Fig. 3(a), four airports are located in illuminated ground areas, which may increase the probability of RFI occurrence. A magnified view of red box A is shown in Fig. 3(b), where the artifacts generated by the RFI overlay in the image are specifically deceptive. Fig. 3(c) shows the magnified view of red box B. The artifacts have strong energy suppression and have already submerged some natural objects and terrain information, seriously affecting the subsequent image interpretation work.

2) Example Case on the Randomness of RFI in SAR Images: Fig. 4 shows an example set of time-series spaceborne SAR images illuminating parts of Riyadh, the capital of Saudi Arabia. Fig. 4(a) is an image of Sentinel-1 acquired on November 7, 2020. It is clear that RFI causes no artifacts. Fig. 4(b) is an image acquired on February 23, 2021. It can be clearly observed that some artifacts are present. Fig. 4(c) is an image acquired on March 7, 2021, with a small number of artifacts. Fig. 4(d) is an image acquired on March 19, 2021, in which the RFI generates many artifacts. All four images illuminate the same ground area, but the location and intensity of the RFI vary due to different acquisition times.

Based on the above properties of RFI in SAR images, using the time correlation and complementary information between the revisited data of the spaceborne SAR system, the intensity of RFI in the background image obtained by stacking and averaging time-series images will be weakened. When the number of images is large enough, an approximately RFI-free background can be obtained, and the difference between the background and the image to be detected forms the basis of the proposed method.

III. Related Works

A. Log-Ratio Operator

The log-ratio operator is a logarithmic operation on the ratio difference image of two images [46]. After logarithmic transformation, the difference image pixels are nonlinearly shrunk, and the contrast between the changed and unchanged classes is enhanced. The nature of the logarithmic operation itself can reduce the significant difference caused by the ratio operation and further reduce the wild point's influence in the background part of the unchanged class, which is more effective when the changed area is smaller than the unchanged area.

Here, the log-ratio operator is used to obtain the difference images between the background and the images to be detected. Its expression is

$$D_u(i,j) = \log\left(\frac{I_u(i,j)}{I_b(i,j)}\right) \tag{6}$$

where D_u represents the generated difference image, I_b and I_u represent the background and the images to be detected, respectively; $u = 1, 2, \ldots, n$. i and j represent the rows and columns of the matrix, respectively.

B. Otsu Algorithm

The Otsu algorithm, also known as the maximum between-class variance method, is an adaptive threshold segmentation algorithm [47]. It divides the image into two parts, background and target, according to the grayscale characteristics of the image or finds a threshold t and splits the color of the image into two parts, $1, 2, \ldots, t$ and $t+1, \ldots, 256$. The principle of algorithm classification is to maximize the class variance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

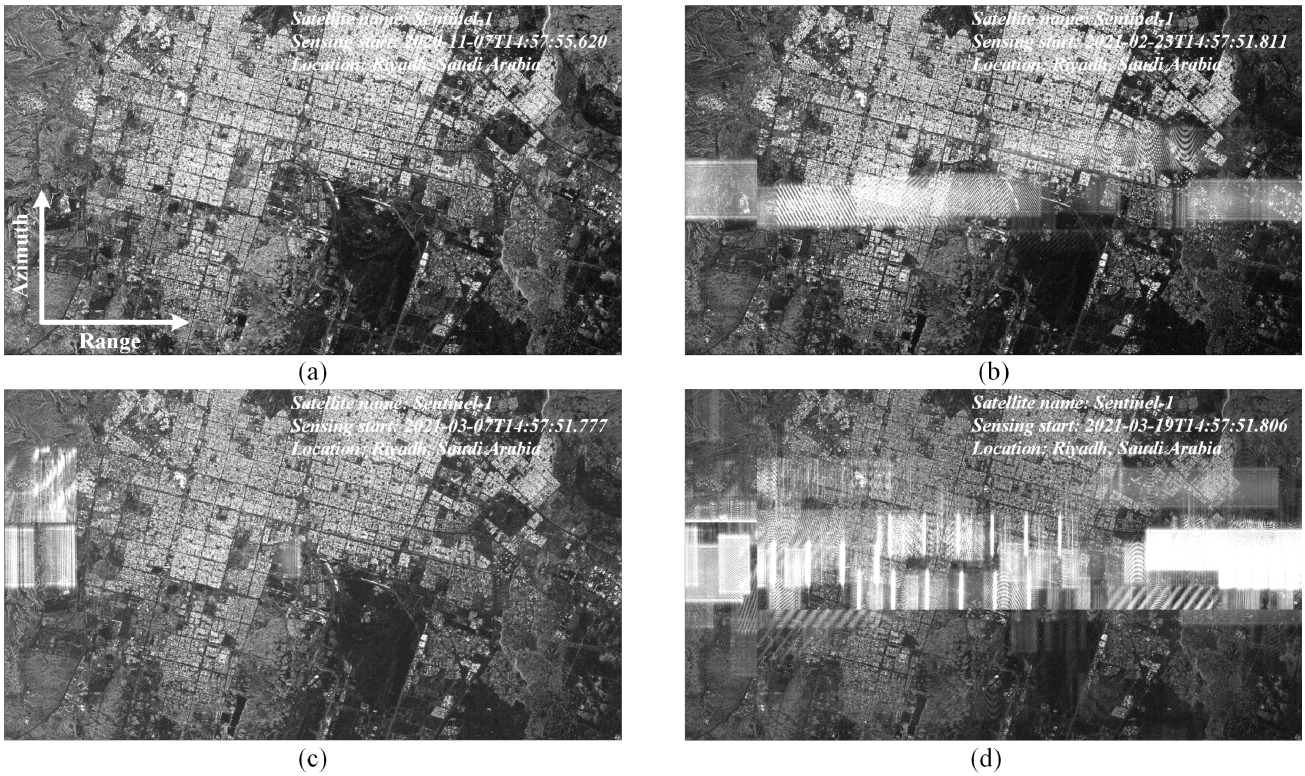


Fig. 4. Example case of time-series SAR images. (a) SAR image without RFI acquired on November 7, 2020. (b) SAR image with RFI acquired on February 23, 2021. (c) SAR image with RFI acquired on March 7, 2021. (d) SAR image with RFI acquired on March 19, 2021.

between the background and the target and to determine the optimal threshold according to the maximum distance between classes [48], [49]. Assuming that image I has L gray levels in total, and there are n pixels with gray level k, the total pixels N of the image I are

$$N = \sum_{k=1}^{L} n_k. \tag{7}$$

The probability P_k of the occurrence of a pixel with gray level k is

$$P_k = \frac{n_k}{N} \tag{8}$$

clearly, $P_k \geq 0$, and

$$\sum_{k=1}^{L} P_k = 1. \tag{9}$$

Assuming the image is divided into two regions with the gray value t as the threshold, the pixel region A (background class) with pixel gray level $1, 2, \ldots, t$, and the pixel gray level $t+1, \ldots, L$ Pixel area B (target class), then the probability P_A and P_B of A and B appearing are

$$P_A = \sum_{k=1}^{t} P_k \tag{10}$$

$$P_B = \sum_{k=t+1}^{L} P_k = 1 - P_A. \tag{11}$$

The grayscale mean values ω_A and ω_B of A and B are

$$\omega_A = \sum_{k=1}^{t} k P_k / P_A \tag{12}$$

$$\omega_B = \sum_{k=t+1}^{L} k P_k / P_B. \tag{13}$$

And the grayscale mean ω_I of image I is

$$\omega_I = P_A \omega_A + P_B \omega_B = \sum_{k=1}^{L} k P_k. \tag{14}$$

From this expression, the between-class variance of the two regions A and B can be obtained as

$$\theta^2(t) = P_A(\omega_A - \omega_I)^2 + P_B(\omega_B - \omega_I)^2. \tag{15}$$

Clearly, P_A, P_B, ω_A, ω_B, ω_I, and $\theta^2(t)$ are all functions of gray level t. To obtain the optimal segmentation threshold, Otsu takes the between-class variance of the two classes as the criterion. It considers that $\tilde{t}$, which maximizes the value of $\theta^2(t)$, is the optimal threshold

$$\tilde{t} = \operatorname{Arg} \operatorname*{Max}_{1 \leq t \leq L} \left[P_A(\omega_A - \omega_I)^2 + P_B(\omega_B - \omega_I)^2 \right]. \tag{16}$$

Because variance measures the uniformity of the gray distribution, the larger the variance is, the greater the difference between the two parts of the image. When part of the target is mistakenly classified as the background or the converse, the difference between the two parts decreases. Therefore, maximizing the between-class variance means minimizing the probability of misclassification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

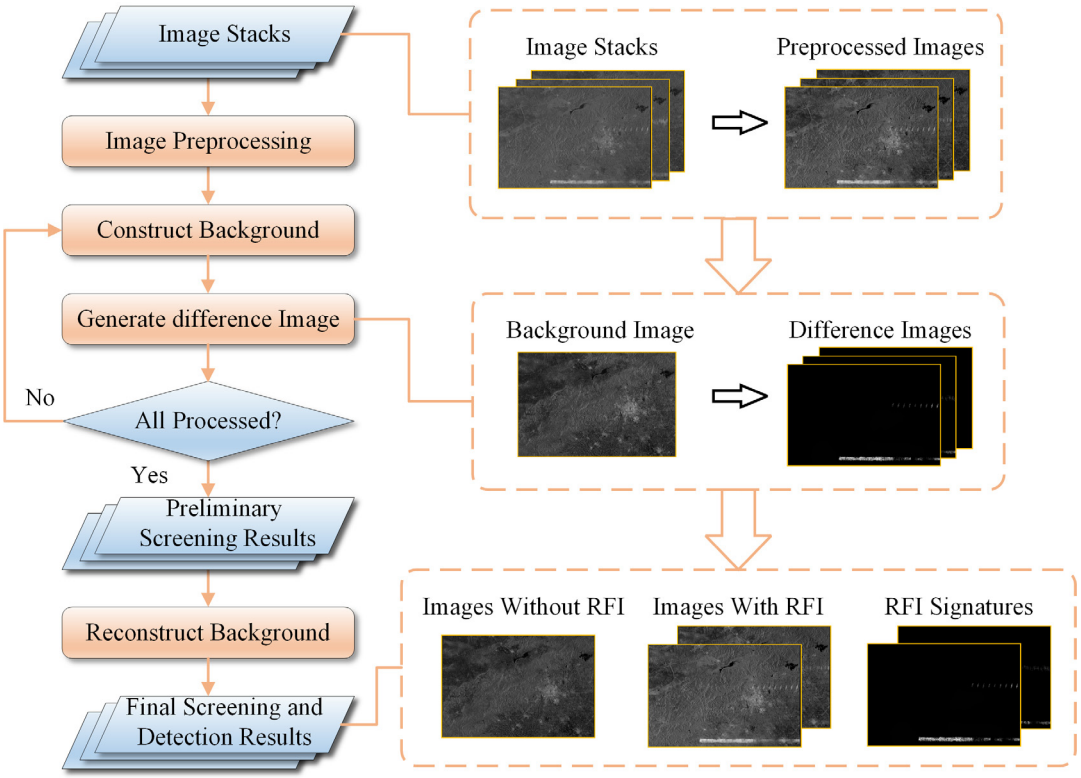


Fig. 5. Flowchart of the proposed method.

C. Image Entropy

The information entropy of an image can represent the distribution characteristics of the image's gray level and indicate the image's texture complexity. It is an important indicator to measure the richness of the image information [50]. According to the principle of Shannon's information theory, the definition of image entropy is as follows:

$$E = -\sum_{k=1}^{L} P_k \log_2(P_k) \tag{17}$$

where E represents the image information entropy, L is the total gray level of the image, and P_k represents the probability of the occurrence of a pixel with a gray value of k. The more gray levels there are in the image, the richer the corresponding texture information, the greater the image information entropy, and by contrast, the smaller the image information entropy.

IV. Methodology

To screen the RFI-containing parts and detect RFI signatures from SAR images quickly and accurately, based on the characteristics of RFI, a method to screen and detect RFI from massive SAR images is proposed in this section. The flowchart of the proposed method is given in Fig. 5.

A. Image Preprocessing

The proposed method uses time-series images to construct the background image, and a preprocessing step is necessary to make the images comparable in spectral and spatial domains [51]. The specific operations are as follows.

1) Radiometric Correction: Since Sentinel-1 GRD products were not radiometrically corrected, significant radiometric deviations were still present. To obtain accurate results for subsequent processing, it is necessary to perform this step on the time-series images so that each pixel value of the image represents the true radar backscatter [52].

2) Image Registration: Since the acquisition time of time-series SAR images has a specific interval, and the satellite orbit deviation will lead to the incomplete coincidence of the images. When the error of the two images is greater than or equal to one pixel, the correlation will be lost entirely [53]. Therefore, it is necessary to register the detected images to match all SAR images to the same spatial scene architecture.

B. Preliminary Screening of RFI

After preprocessing, the time-series images are stacked and averaged to obtain an approximately RFI-free background image. The change detection method is applied to all the images to be detected and the background image, which is primarily divided into two steps: generating and analyzing the difference image. The detailed flowchart is shown in Fig. 6.

1) Generation of Difference Images: The generation of the difference image is actually intended to find a matrix that can represent the distance between two SAR images and visualize the matrix. Since SAR speckle noise is multiplicative noise, the log-ratio operation is more suitable for the statistical characteristics of SAR images. It has the advantages of low computational complexity and easy implementation. Therefore, an improved log-ratio operator is selected for use in this

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

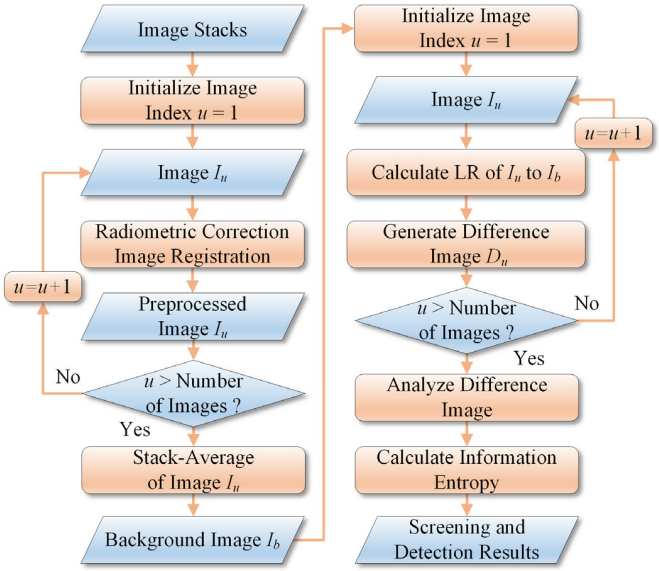


Fig. 6. Flowchart for preliminary detection of RFI.

article, and its expression is

$$D_u(i,j) = \log\left(\frac{I_u(i,j)+1}{I_b(i,j)+1}\right) \tag{18}$$

where D_u represents the difference image, I_b and I_u represent the background and the images to be detected. Compared with the traditional log-ratio operator, the addition of the constant to the formula prevents the occurrence of infinity. When $D_u < 0$, it means that there is a strong point in I_b, the gray value of the pixel in I_u is lower or zero, and there is no RFI in I_u at this time. Before using the Otsu method, we set the D_u value of this pixel to zero to avoid false alarms.

2) Analysis of Difference Image: The analysis methods of difference images commonly used in SAR image change detection include threshold analysis, cluster analysis, graph-cut analysis, and level-set analysis. The threshold method divides the difference image into two categories based on the threshold, which has the advantages of high efficiency and simplicity. This article uses the Otsu method to obtain the adaptive segmentation threshold t of the difference image to eliminate false alarm information. The operation of segmenting the difference image can be defined as

$$\tilde{D}_u(i,j) = \begin{cases} 1, & D_u(i,j) \geq \tilde{t}_u \\ 0, & D_u(i,j) < \tilde{t}_u \end{cases} \tag{19}$$

where D_u represents the difference image before threshold segmentation, and $\tilde{D}_u$ represents the difference image after adaptive segmentation. After the above operations are performed on the difference images, the speckle noise in the binary image is eliminated based on the morphological filtering method [54]. Since RFI shows strong energy in the images, the presence of RFI is greatly weakened in the construction process of the background, and the remaining ground object information is preserved. Therefore, after the operations mentioned above, the remaining part of the difference image is the RFI information. Since the entropy of the difference image with RFI is much larger than that without RFI, we calculate the entropy of all difference images and use the K-means clustering method to classify them, which can obtain the preliminary screening results of SAR images.

C. Secondary Screening and Detection of RFI

After the preliminary detection of RFI was completed in the time-series images, the above results were used to exclude the part of the images containing RFI, the background is reconstructed using the parts without RFI. This approach replaces the approximately RFI-free background in the preliminary detection work, which can effectively improve the RFI detection accuracy. Based on the reconstructed background, the final SAR image screening and RFI detection results can be obtained by applying the method proposed in this article to all the images to be screened.

V. Experimental Data Description

To verify the effectiveness of the proposed method, four sets of meaningful and challenging datasets are tested. The dataset used in the experiment is composed of GRD products obtained by Sentinel-1, which are located in Beijing, Kaifeng, Hangzhou, and parts of the Arabian Peninsula. The images were acquired between January 27, 2019, and March 31, 2021, with a detection interval of 12 days. The size of a single image is approximately 25 000 × 16 000, and the details of the dataset are shown in Table I. The geographical locations of these products are shown in Fig. 7. Some of the pseudocolor images with RFI are shown in Fig. 8. The RFI forms in these images vary and contain complex scenes, such as cities, vegetation, and bare ground.

VI. Experimental Results and Discussion

In this section, the experimental results on the four regional datasets obtained by the proposed method are presented and analyzed. The potential application of this method is discussed based on the experimental results and related analytical work.

A. Example #1: Sentinel-1 Images of Beijing Surrounding Area With RFI

In the first example, a dataset containing 30 Sentinel-1 time-series images is used for the experiment, which were acquired from March 29, 2020, to March 12, 2021. The illuminated ground area is around Beijing city, and the size of a single image is 25 670 × 16 732. Among them, 15 images obtained before September 20, 2020, were used to construct the background, and the rest were used as the test set. All the images in this dataset are first preprocessed. After the background is constructed, the method proposed in this article is used for preliminary screening, and the background is reconstructed after removing some images containing RFI. Lastly, the images to be detected in the test set are subjected to secondary screening to obtain the final result. Fig. 9 shows partial screening and RFI detection results. Fig. 9(a) is the background image obtained by applying the proposed method, and it can be observed that there are no RFI artifacts in the background. Fig. 9(b) and (f) are images containing RFI in time series, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

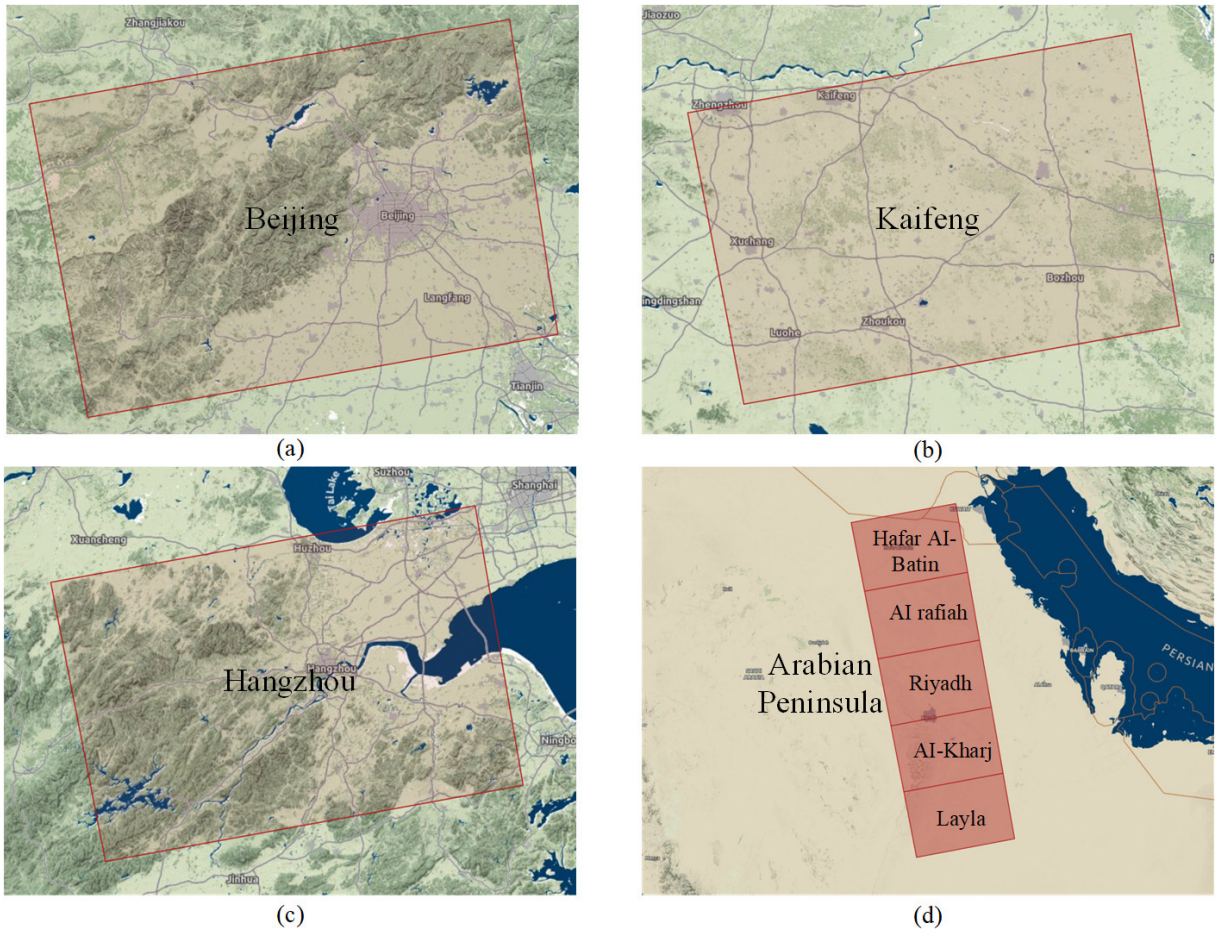


Fig. 7. Geolocations of the experimental dataset. (a) Surrounding areas of Beijing. (b) Surrounding areas of Kaifeng. (c) Surrounding areas of Hangzhou. (d) Five contiguous areas of Hafar Al-Batin, Al Rafiah, Riyadh, Al-Kharj, and Layla in the Arabian Peninsula.

TABLE I
DETAILS OF THE EXPERIMENTAL DATASET

Geolocations		Number of Images	Size of Images	Sensing Period	Images to Construct The Background	Image Used for Verification
China	Beijing	30	25711×16717	2020/3/29-2021/3/12	2020/3/29-2020/9/13	2020/9/25-2021/3/12
	Kaifeng	30	25670×16732	2019/1/27-2020/1/10	2019/1/27-2019/7/14	2019/7/26-2020/1/10
	Hangzhou	30	25680×16720	2020/3/24-2021/3/7	2020/3/24-2020/9/8	2020/9/20-2021/3/7
Arabian Peninsula	Hafar Al-Batin	30	25787×16736	2020/4/17-2021/3/31	2020/4/17-2020/10/2	2020/10/14-2021/3/31
	Al rafiah	30	25780×16736	2020/4/17-2021/3/31	2020/4/17-2020/10/2	2020/10/14-2021/3/31
	Riyadh	30	25779×16736	2020/4/17-2021/3/31	2020/4/17-2020/10/2	2020/10/14-2021/3/31
	Al-Kharj	30	25768×16737	2020/4/17-2021/3/31	2020/4/17-2020/10/2	2020/10/14-2021/3/31
	Layla	30	25763×16736	2020/4/17-2021/3/31	2020/4/17-2020/10/2	2020/10/14-2021/3/31

it is evident that the intensity and position of RFI in different images have certain randomness. Fig. 9(c) shows the difference image after the log-ratio operation. It can be seen from the enlarged yellow box that most of the point targets in the image to be detected are eliminated by the log-ratio operator with the background, thus avoiding the occurrence of false alarms to a certain extent. Fig. 9(d) is the result after using the Otsu method, and the residual point target in Fig. 9(c) is further eliminated. Fig. 9(d) and (g) are the detection results of RFI artifacts corresponding to Fig. 9(b) and (f). The results show that the method can quickly complete the screening of images containing RFI and detect the position of RFI in the image.

In this example, all the images in the dataset are batch-processed using the method proposed in this article. By calculating the entropy of all difference images separately, and the K-means method is used to classify them, the RFI-contaminated images can be screened out. As shown in Fig. 10, among the 30 images in the areas surrounding Beijing, 24 images were found to contain RFI. Eleven of the images for the initial background construction have RFI (as indicated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

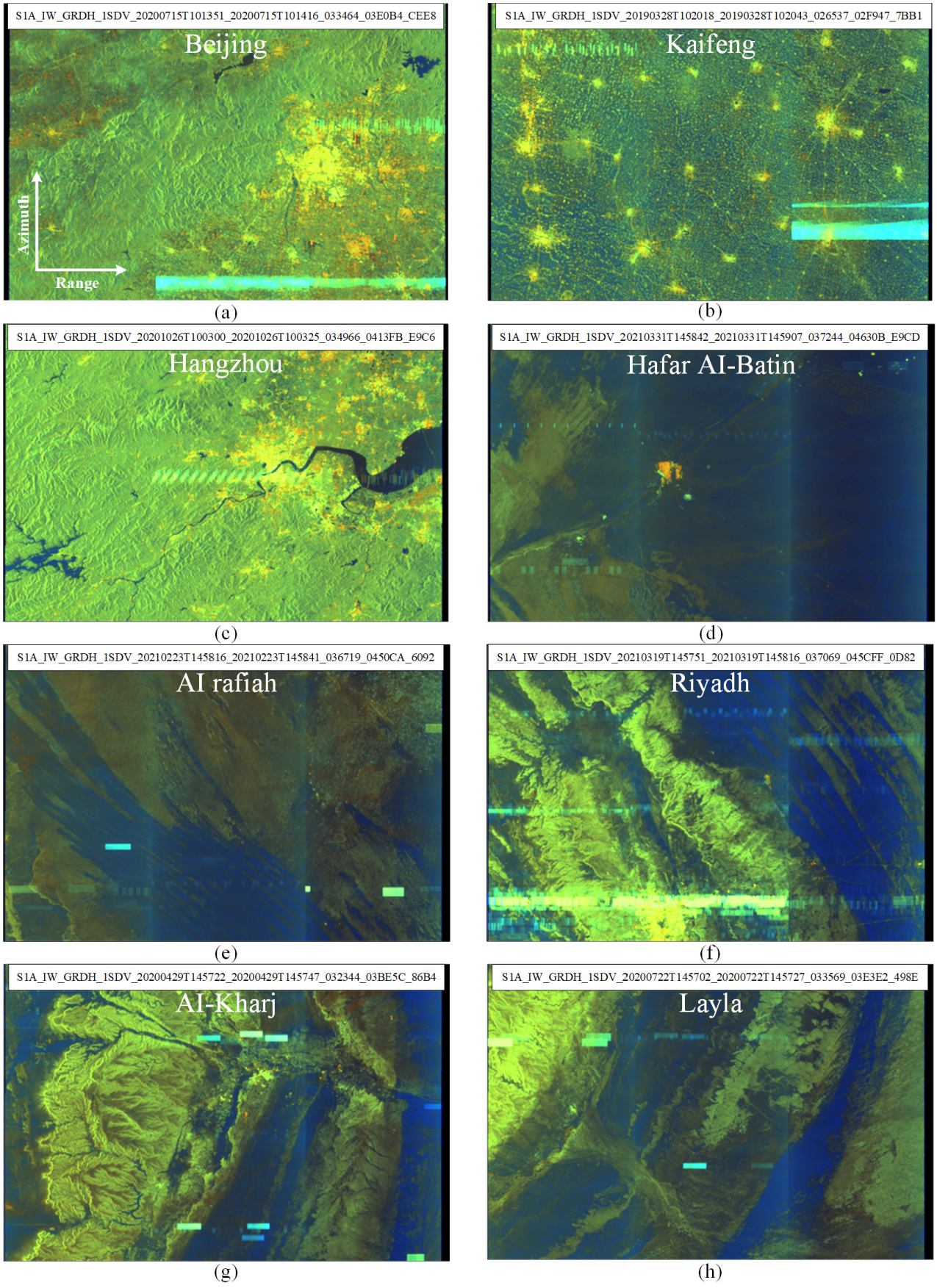


Fig. 8. Pseudocolor images acquired in the experimental area. (a) Surrounding areas of Beijing. (b) Surrounding areas of Kaifeng. (c) Surrounding areas of Hangzhou. (d) Surrounding areas of Hafar Al-Batin. (e) Surrounding areas of Al Rafiah. (f) Surrounding areas of Riyadh. (g) Surrounding areas of Al-Kharj. (h) Surrounding areas of Layla.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

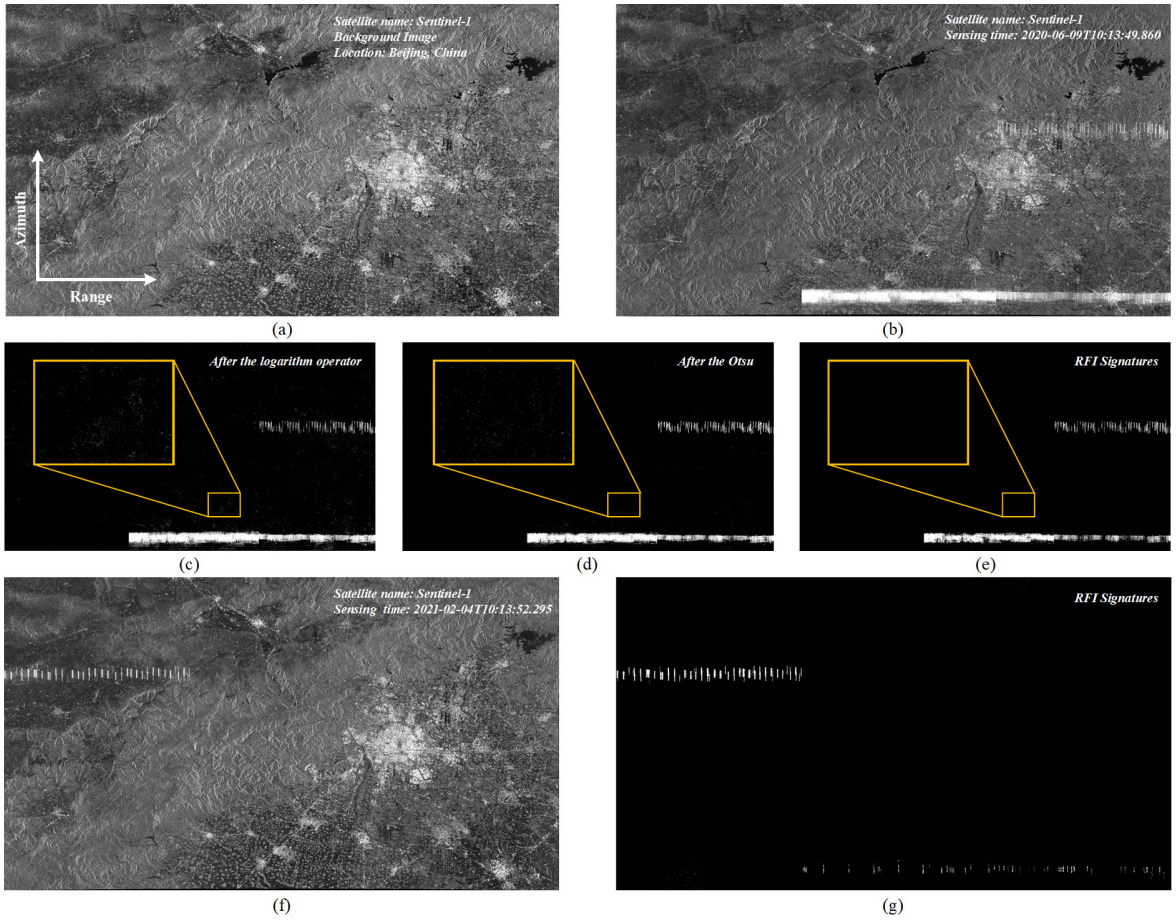


Fig. 9. Experimental results of example #1. (a) Background image. (b) Sentinel-1 image acquired on June 9, 2020. (c) Difference image after the log-ratio operation. (d) Difference image after the Otsu method. (e) and (g) RFI signatures. (f) Sentinel-1 image acquired on February 4, 2021.

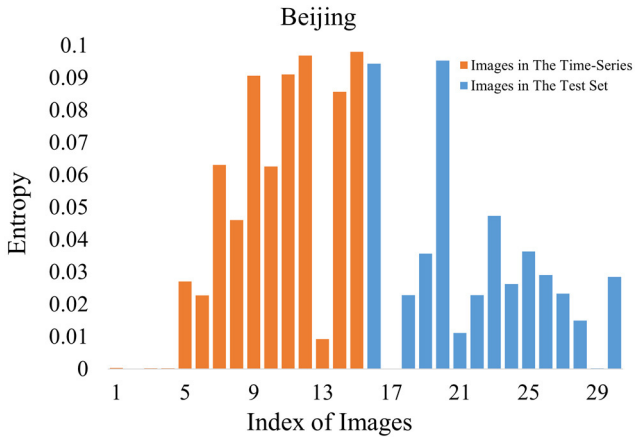


Fig. 10. Screen results of example #1 obtained using the proposed method for all the images in the experimental dataset.

by the orange bars), and RFI contaminates 13 of the 15 images used as the test set (as indicated by the blue bars).

B. Example #2: Sentinel-1 Images of Kaifeng Surrounding Area With RFI

In the second example, a dataset containing 30 Sentinel-1 time-series images is used for the experiment, which were acquired from January 27, 2019, to January 10, 2020. The illuminated ground area is around Kaifeng city, Henan province, and the size of a single image is 25 670 × 16 732. Among them, 15 images obtained before July 14, 2019, were used to construct the background, and the rest were used as the test set. All the images in this dataset are first preprocessed for radiometric correction and image registration. After the background is constructed, the method proposed in this article is used for preliminary screening. According to the preliminary screening results, the background is reconstructed to complete the secondary screening and detection. Fig. 11 shows partial screening and RFI detection results. Fig. 11(a) is the background image obtained by applying the proposed method, and it is clear that there are no RFI artifacts in the background. Fig. 11(b) is an image containing RFI in the time series. Fig. 11(c) shows the corresponding RFI artifacts detected by the proposed method. Fig. 11(d) and (f) are the images containing RFI in the test set. Fig. 11(e) and (g) are the corresponding RFI artifacts, and it can be clearly observed that the intensity and position of the RFI in images acquired at different times have a given randomness. As shown in the results, the proposed method can quickly complete the screening of images containing RFI and accurately detect the position of RFI in the image.

In this example, all the images in the dataset are batch-processed using the method proposed in this article. By calculating the entropy of all difference images separately, and the K-means method is used to classify them, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

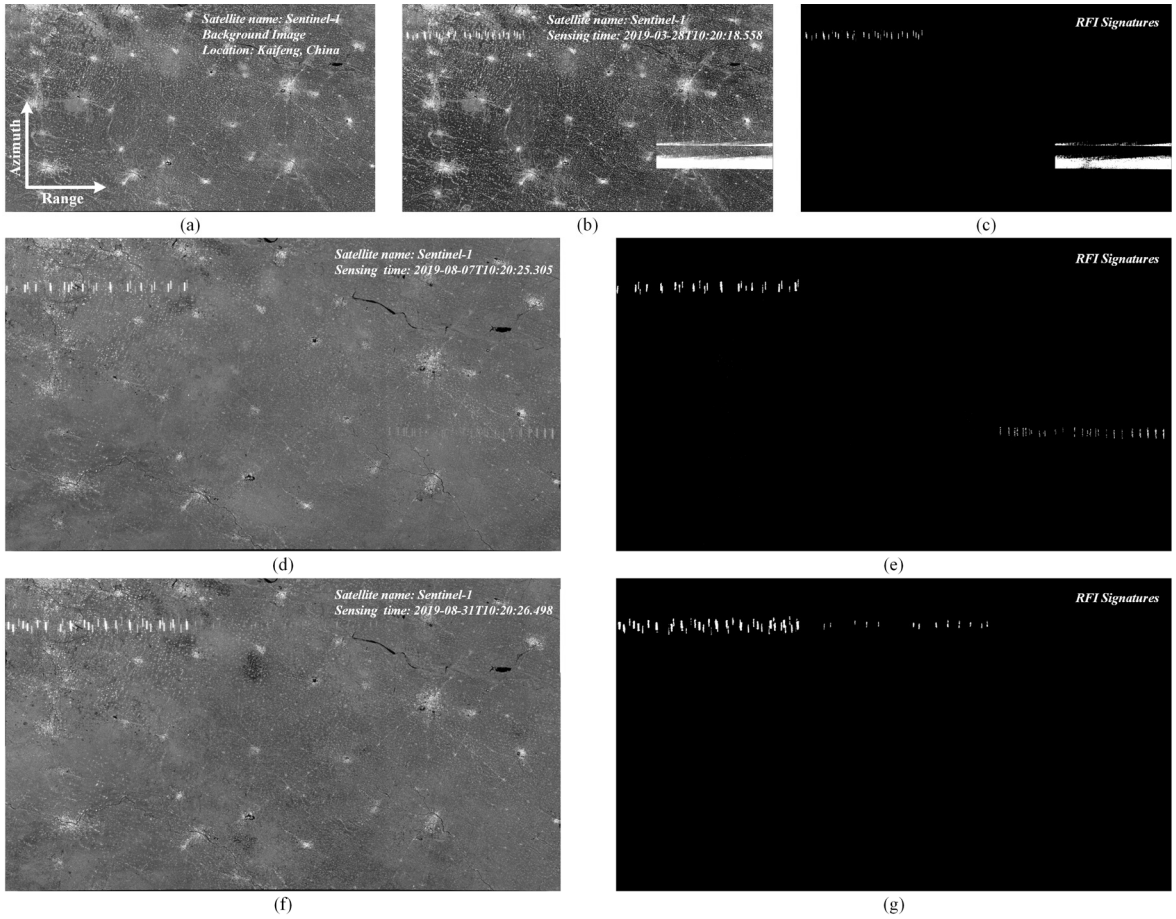


Fig. 11. Experimental results of example #2. (a) Background image. (b) Sentinel-1 image acquired on March 28, 2019. (c), (e), and (g) RFI signatures. (d) Sentinel-1 image acquired on August 7, 2019. (f) Sentinel-1 image acquired on August 31, 2019.

RFI-contaminated images can be screened out. As shown in Fig. 12, among the 30 images in the areas surrounding Kaifeng, 15 images were found to contain RFI. Ten of the images for the preliminary background construction have RFI (as indicated by the orange bars), and RFI contaminates 5 of the 15 images used as the test set (as indicated by the blue bars).

C. Example #3: Sentinel-1 Images of Hangzhou Surrounding Area With RFI

In the third example, a dataset containing 30 Sentinel-1 time-series images is used for the experiment, which were acquired from March 24, 2020, to March 7, 2021. The illuminated ground area is around Hangzhou city, Zhejiang province, and the size of a single image is 25 680 × 16 720. Among them, 15 images obtained before September 20, 2020, were used to construct the background, and the rest were used as the test set. All the images in this dataset are first preprocessed for radiometric correction and image registration. After constructing the background, the proposed method is used for the preliminary screening and detection of RFI. Since all the images used to construct the background contain RFI in this example, the images containing RFI cannot be eliminated to reconstruct the background and perform secondary screening and detection. Fig. 13 shows partial screening and RFI detection results. Fig. 13(a) is the background

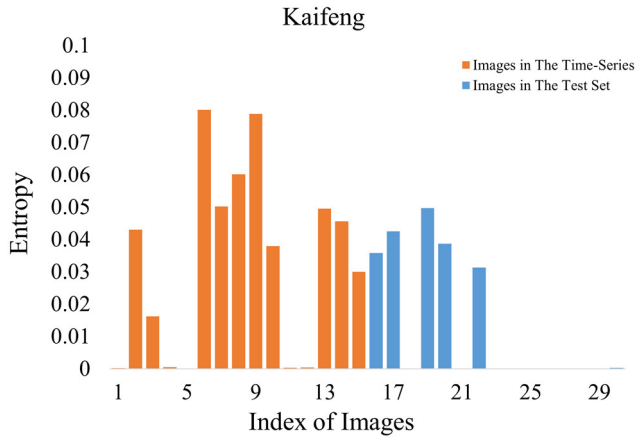


Fig. 12. Screen results of example #2 obtained using the proposed method for all the images in the experimental dataset.

image obtained by applying the proposed method. Due to the presence of RFI in all the images, the background has residual RFI artifacts. Fig. 13(b) is an image containing RFI in the time series. Fig. 13(c) shows the corresponding RFI artifacts detected by the proposed method. Fig. 13(d) and (f) are the images containing RFI in the test set. Fig. 13(e) and (g) are the corresponding RFI artifacts. As shown in the images of the detection results, a completely RFI-free background cannot be obtained in the case of continuous RFI in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

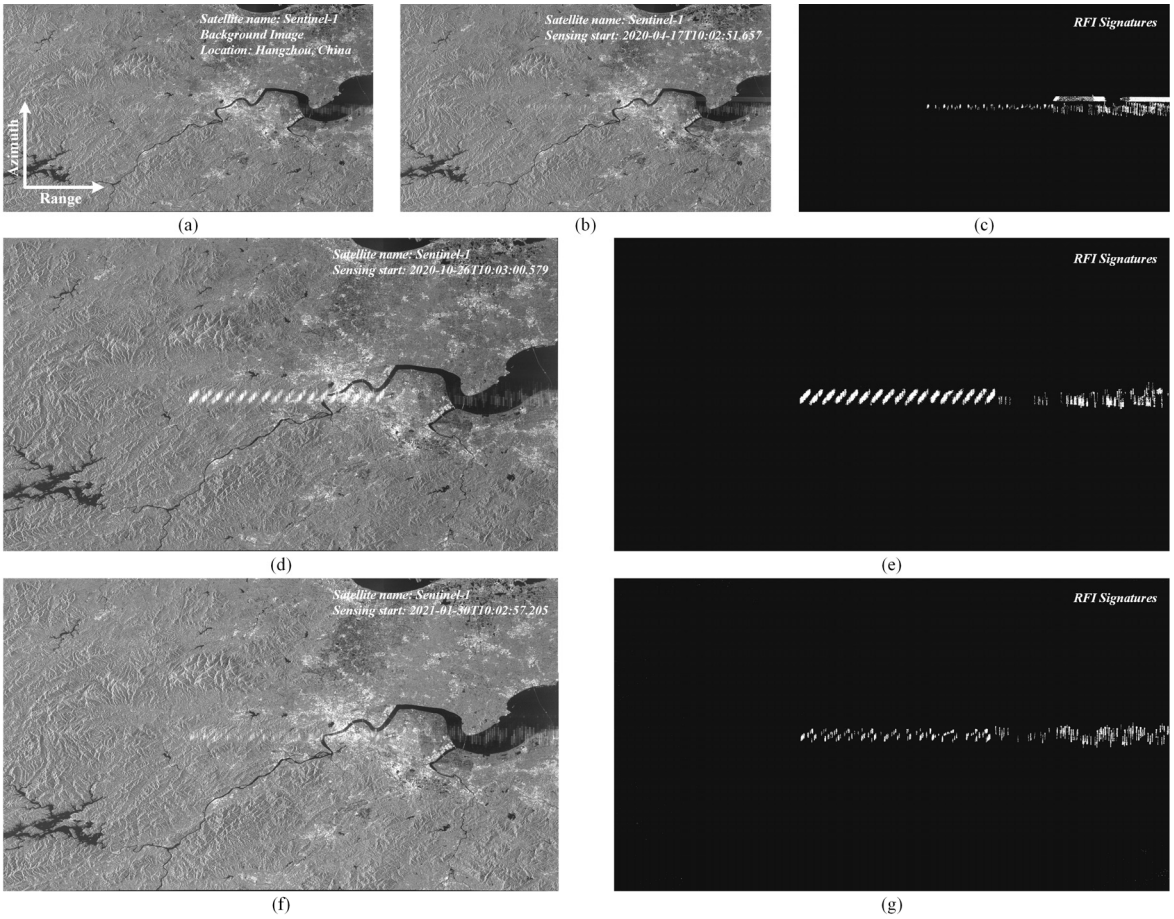


Fig. 13. Experimental results of example #3. (a) Background image. (b) Sentinel-1 image acquired on April 17, 2020. (c), (e), and (g) RFI signatures. (d) Sentinel-1 image acquired on October 26, 2019. (f) Sentinel-1 image acquired on January 30, 2019.

same area for all images in the dataset. However, due to the characteristics of RFI in SAR images, the more images the dataset uses, the weaker the RFI becomes in the background. Therefore, the proposed method can still screen the images with RFI and detect the location of the RFI in the image.

In this example, all the images in the dataset are batch-processed using the method proposed. By calculating the entropy of all difference images separately, as shown in Fig. 14, and the K-means method is used to classify them, the RFI-contaminated images can be screened out. Among the 30 images were screened out containing RFI. Unlike Example #1 and Example #2, two RFI-contaminated images in this example were missed due to residual RFI in the background and weak RFI power in the image to be detected. This shows that the proposed method has limitations when facing the situation that all images contain RFI in the same scene.

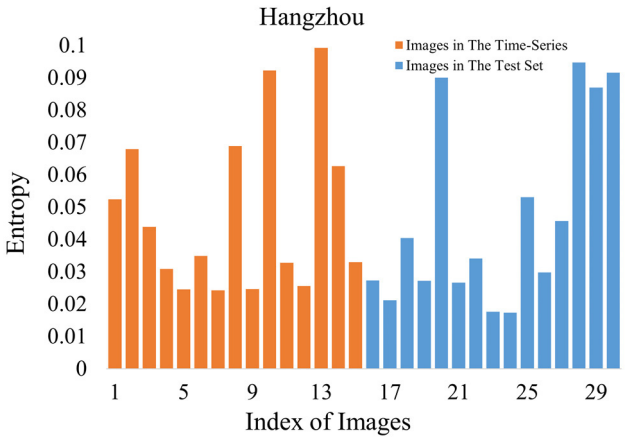


Fig. 14. Screen results of example #3 obtained using the proposed method for all the images in the experimental dataset.

D. Example #4: Sentinel-1 Images of Arabian Peninsula Area With RFI

In the fourth example, a dataset containing 150 Sentinel-1 time-series images from five consecutive regions of the Arabian Peninsula is used for the experiments, and they were acquired from April 17, 2020, to March 31, 2021. The size of a single image is 25 780 × 16 736. The images obtained before September 20, 2020, were used to construct the background, and the rest were used as the test set. Fig. 15 shows the screening results of some images containing RFI. Fig. 15(a), (d), (g), (j), and (m) show the preprocessed images containing RFI in the test set, from which it can be seen that RFI produces some artifacts, and there is an apparent radiation inconsistency in the position of image stitching. Fig. 15(b), (e), (h), (k), and (n) show the background images constructed using the proposed method. It can be seen that most of the RFI-generated artifacts are already invisible. Some RFIs remain in Fig. 15(h) because

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

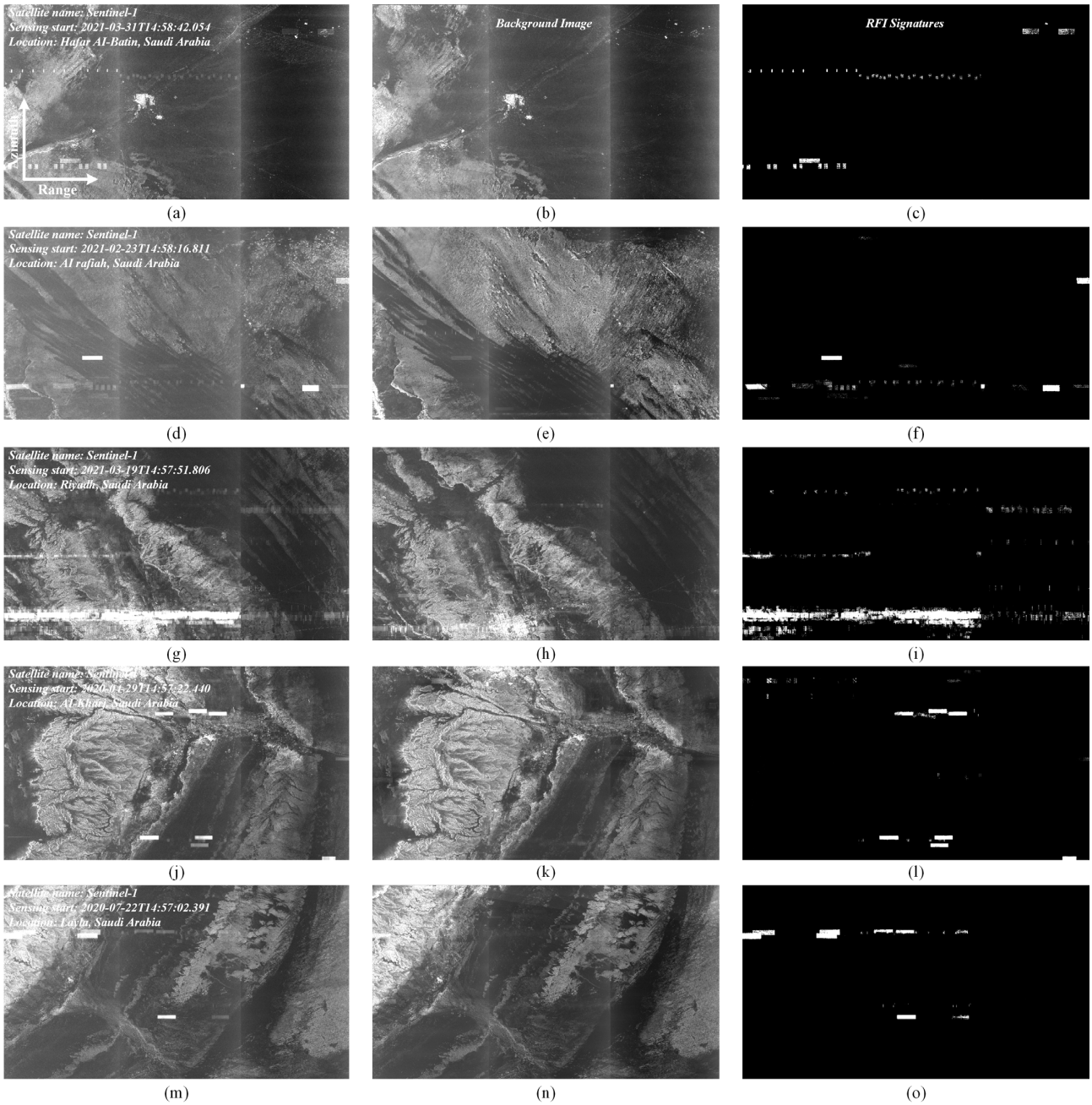


Fig. 15. Experimental results for example #4. (a) Sentinel-1 image with RFI from the area around Hafar Al-Batin. (b), (e), (h), (k), and (n) Background image. (c), (f), (i), (l), and (o) RFI detection results. (d) Sentinel-1 image with RFI from the area around Al Rafiah. (g) Sentinel-1 image with RFI on the area around Riyadh. (j) Sentinel-1 image with RFI on the area around Al-Kharj. (m) Sentinel-1 image with RFI on the area around Layla.

most of the time-series images in this region contain RFIs. Fig. 15(c), (f), (i), (l), (o) are the detection results of the RFIs corresponding to Fig. 15(a), (d), (g), (j), and (m), It can be seen from Fig. 15(i) that due to the influence of the background, the detection results contain some false alarms.

As shown in the detection results, the proposed method can still accurately screen for the presence or absence of RFI from the image in the face of complex terrain and multiple forms of RFI images.

Batch screening is performed using the method proposed on the dataset. The images contaminated by RFI can be quickly screened by separately calculating the entropy of the obtained difference images. As shown in Fig. 16, among the 150 images in the five regions of Hafar Al-Batin, Al rafiah, Riyadh, Al-Kharj, and Layla, the numbers of images containing RFI are 6, 18, 27, 22, and 20, respectively.

E. On the Performance of RFI Screening and Detection

To quantitatively evaluate the performance of the proposed method in the screening of RFI-containing images, the screening accuracy is obtained by comparing all the screening results with the actual RFI-containing images. Furtherly, to analyze the performance of the proposed method in the detection of RFI artifacts, three quality evaluation indices for all RFI-containing images, i.e., PR, RR, and $F1$-score, are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

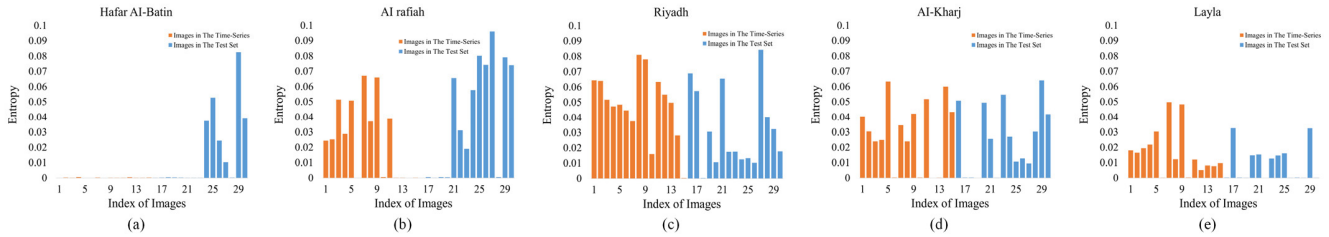


Fig. 16. Screen results for example #4 obtained using the proposed method for all the images in the experimental dataset. (a) Surrounding areas of Hafar Al-Batin. (b) Surrounding areas of Al Rafiah. (c) Surrounding areas of Riyadh. (d) Surrounding areas of Al-Kharj. (e) Surrounding areas of Layla.

TABLE II

QUANTITATIVE EVALUATION INDICES OF THE EXPERIMENTAL DATASET

Scene	Size of Images	Precision of Screening (%)	PR (%)	RR (%)	F1-Score (%)
Beijing	25711×16717	100.00	92.86	90.24	91.53
Kaifeng	25670×16732	100.00	93.53	91.68	92.60
Hangzhou	25680×16720	93.33	91.18	87.56	89.33
Hafar Al-Batin	25787×16736	100.00	93.71	91.79	92.74
Al rafiah	25780×16736	100.00	92.25	90.37	91.30
Riyadh	25779×16736	96.67	90.04	86.62	88.30
Al-Kharj	25768×16737	100.00	92.61	90.85	91.72
Layla	25763×16736	100.00	92.59	91.26	91.92

calculated and averaged in Table II. The label information with and without RFI in the image is manually annotated, and each label occupies 2500 pixels. The PR is defined as the ratio of correctly detected RFIs among all RFI tags, which can be expressed as

$$\mathrm{PR}(\mathrm{TP},\mathrm{FP}) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{20}$$

where the TP denotes the true positives, i.e., the pixels of the RFI that are correctly detected. FP denotes the false positive, i.e., the pixels of RFI incorrectly detected (false alarm).

The RR is a measure of coverage, and it is defined as the ratio of the RFI being detected in all ground truths, which can be expressed as

$$\mathrm{RR}(\mathrm{TP},\mathrm{FN}) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{21}$$

where FN denotes the false negatives, and it can be understood as the missed detection.

The $F1$-score is defined as a harmonic mean between PR and RR, which can be expressed as

$$F1-\mathrm{score}(\mathrm{PR},\mathrm{RR}) = 2\times\frac{\mathrm{PR}\times\mathrm{RR}}{\mathrm{PR}+\mathrm{RR}}. \tag{22}$$

It can be seen from Table II that the proposed method has a good performance in screening and detection. In terms of screening accuracy, the six regions of Beijing, Kaifeng, Hafar Al-Batin, Al Rafiah, Al-Kharj, and Layla can accurately screen out images with RFI. However, missed screening occurs due to the residual RFI artifacts in the background images of Hangzhou and Riyadh. In terms of the detection accuracy of RFI artifacts, the RR of the proposed method in all experimental areas is higher than 90%, which shows that the method can accurately detect RFI artifacts while reducing the impact of false alarms. The PRs in the table reflect the missed detection of RFI artifacts, and it can be seen that all regions have PRs above 85%. Compared with Hangzhou and Riyadh regions, where only primary screening was used, the PRs of the other six regions used secondary screening and detection were significantly higher. This shows that using a secondary screening step can effectively reduce the occurrence of RFI missed detection.

F. Discussion

For many years, conventional RFI detection methods for SAR products have been based on Level-1 products because these methods have relied on spectrum analysis. This article details a simple RFI detection method that can perform batch operations on a large number of SAR Level-2 products, which makes the proposed method more valuable for practical engineering. Although batch RFI detection work has only completed experiments in a few regions, the proposed method provides new opinions for implementing more meaningful tasks.

1) Over time, a global RFI-free background database of Sentinel-1 can be generated and updated. In this way, it is possible to detect whether the newly acquired image products contain interference batch-by-batch effectively.
2) It can provide a robust and efficient approach for Sentinel-1 data quality assessment and RFI detection, which are essential for specific applications requiring time series imagery products, such as agricultural surveys or natural disaster assessments. Images heavily contaminated by RFI can be detected by the proposed method and discarded automatically without time-consuming manual inspection.

However, since the proposed method is based on SAR image change detection technology, when there are more moving strong point targets (such as maritime ships) in the scene, false detection of RFI will occur.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

VII. Conclusion

In this article, a novel approach to screen and detect RFI simultaneously from massive SAR images by constructing an RFI-free background is proposed, which provides a new view for RFI detection in the image domain. Unlike traditional RFI detection methods, this approach does not require prior knowledge of whether SAR images contain RFI. It involves the screening and detection of RFI by comparing the difference between the RFI-free background and the images need to be detected. Experimental results based on Sentinel-1 images show that the method has good performance and is easy to implement in engineering. It is expected to play an important role in the batch screening of RFI-containing images and the assessment of SAR data quality.

Acknowledgment

The authors would like to thank the European Space Agency (ESA) for providing the Sentinel-1 data free online.

References

[1] A. Reigber *et al.*, "Very-high-resolution airborne synthetic aperture radar imaging: Signal processing and applications," *Proc. IEEE*, vol. 101, no. 3, pp. 759–783, Mar. 2013.

[2] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[3] Y. Deng, W. Yu, H. Zhang, W. Wang, D. Liu, and R. Wang, "Forthcoming spaceborne SAR development," *J. Radars*, vol. 9, no. 1, pp. 1–33, Feb. 2020.

[4] F. Zhou and M. Tao, "Research on methods for narrow-band interference suppression in synthetic aperture radar data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3476–3485, Jul. 2015.

[5] H. Yang, C. Chen, S. Chen, F. Xi, and Z. Liu, "SAR RFI suppression for extended scene using interferometric data via joint low-rank and sparse optimization," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 11, pp. 1976–1980, Nov. 2021.

[6] Y.-K. Deng, F.-J. Zhao, and Y. Wang, "Brief analysis on the development and application of spaceborne SAR," *J. Radars*, vol. 1, no. 1, pp. 1–10, 2012.

[7] Q. Wang, X. Zhang, G. Chen, F. Dai, Y. Gong, and K. Zhu, "Change detection based on faster R-CNN for high-resolution remote sensing images," *Remote Sens. Lett.*, vol. 9, no. 10, pp. 923–932, Aug. 2018.

[8] P. Washaya, T. Balz, and B. Mohamadi, "Coherence change-detection with Sentinel-1 for natural and anthropogenic disaster monitoring in urban areas," *Remote Sens.*, vol. 10, no. 7, p. 1026, Jun. 2018.

[9] F. D. Traglia, A. Ciampalini, G. Pezzo, and M. Battaglia, "Synthetic aperture radar and natural hazards: Applications and outlooks," *Frontiers Earth Sci.*, vol. 7, p. 191, Jul. 2019.

[10] J. D. Alban, G. Connette, P. Oswald, and E. Webb, "Combined Landsat and L-band SAR data improves land cover classification and change detection in dynamic tropical landscapes," *Remote Sens.*, vol. 10, no. 2, p. 306, Feb. 2018.

[11] J. Montgomery, B. Brisco, L. Chasmer, K. Devito, D. Cobbaert, and C. Hopkinson, "SAR and lidar temporal data fusion approaches to boreal wetland ecosystem monitoring," *Remote Sens.*, vol. 11, no. 2, p. 161, Jan. 2019.

[12] Z. Lv, N. Li, Z. Guo, and J. Zhao, "Detection and mitigation of mutual RFI in C-band SAR: A case study of Chinese GaoFen-3," in *Proc. IEEE Radar Conf. (RadarConf21)*, May 2021, pp. 1–5.

[13] Y. Li, D. Huang, S. Xing, and X. Wang, "A review of synthetic aperture radar jamming technique," *J. Radars*, vol. 9, no. 5, pp. 753–764, Oct. 2020.

[14] W. Xu, W. Xing, C. Fang, P. Huang, and W. Tan, "RFI suppression based on linear prediction in synthetic aperture radar data," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 12, pp. 2127–2131, Dec. 2021.

[15] H. Zhang, Y. Huang, J. Li, Z. Chen, L. Cai, and W. Hong, "Time-varying RFI mitigation for SAR systems via graph Laplacian clustering techniques," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[16] Y. Huang, G. Liao, L. Zhang, Y. Xiang, J. Li, and A. Nehorai, "Efficient narrowband RFI mitigation algorithms for SAR systems with reweighted tensor structures," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9396–9409, Nov. 2019.

[17] Y. Huang *et al.*, "HRWS SAR narrowband interference mitigation using low-rank recovery and image-domain sparse regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[18] J. Su, H. Tao, M. Tao, L. Wang, and J. Xie, "Narrow-band interference suppression via RPCA-based signal separation in time-frequency domain," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 11, pp. 5016–5025, Nov. 2017.

[19] H. Yang, K. Li, J. Li, Y. Du, and J. Yang, "BSF: Block subspace filter for removing narrowband and wideband radio interference artifacts in single-look complex SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[20] X. Lu *et al.*, "Radio frequency interference suppression for SAR via block sparse Bayesian learning," *IEEE J. Sel. Topics Signal Process.*, vol. 11, no. 12, pp. 4835–4847, Dec. 2018.

[21] Y. Huang, G. Liao, J. Li, and J. Xu, "Narrowband RFI suppression for SAR system via fast implementation of joint sparsity and low-rank property," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2748–2761, May 2018.

[22] H. Yang, Y. He, Y. Du, T. Zhang, J. Yin, and J. Yang, "Two-dimensional spectral analysis filter for removal of LFM radar interference in space-borne SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[23] 2022 S-1 RFI Maps. (2022). *Aresys Earth Observation*. [Online]. Available: https://s1rfimap.aresys.it

[24] M. Tao, J. Su, Y. Huang, and L. Wang, "Mitigation of radio frequency interference in synthetic aperture radar data: Current status and future trends," *Remote Sens.*, vol. 11, no. 20, p. 2438, Oct. 2019.

[25] Y. Huang, B. Zhao, M. Tao, Z. Chen, and W. Hong, "Review of synthetic aperture radar interference suppression," *J. Radars*, vol. 9, no. 1, pp. 86–106, 2020.

[26] H.-F. Zheng, L. Yang, and J. Feng, "Research on the subband subspace filtering for narrow band interference suppression in SAR," *J. Electron. Inf. Technol.*, vol. 35, no. 12, pp. 2836–2842, Dec. 2013.

[27] N. Li, Z. Lv, Z. Guo, and J. Zhao, "Time-domain notch filtering method for pulse RFI mitigation in synthetic aperture radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] T. Yamamoto, K. Takeda, and F. Adachi, "A study of frequency-domain signal detection for single-carrier transmission," in *Proc. IEEE 70th Veh. Technol. Conf. Fall*, Sep. 2009, pp. 1–5.

[29] H. Liu, D. Li, Y. Zhou, and T. Truong, "Simultaneous radio frequency and wideband interference suppression in SAR signals via sparsity exploitation in time-frequency domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5780–5793, Oct. 2018.

[30] Z. Lv, H. Zhang, N. Li, and Z. Guo, "A two-step approach for pulse RFI detection in SAR data," in *Proc. IEEE Sensors*, Oct. 2021, pp. 1–4.

[31] S. Buckreuss and R. Horn, "E-SAR P-band SAR subsystem design and RF-interference suppression," in *Proc. Sens. Manag. Environ. IEEE Int. Geosci. Remote Sensing. Symp. (IGARSS)*, Jul. 1998, pp. 466–468.

[32] Y. Luo, B. Wen, and H. Zhou, "Suppressing of radio-frequency interferences in chirp radar based on averaging range spectrum," *Wuhan Univ. J. Natural Sci. Ed.*, no. 5, pp. 649–652, May 2003.

[33] F. Zhou, R. Wu, M. Xing, and Z. Bao, "Eigensubspace-based filtering with application in narrow-band interference suppression for SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 1, pp. 75–79, Jan. 2007.

[34] J. Feng, H. Zheng, Y. Deng, and D. Gao, "Application of subband spectral cancellation for SAR narrow-band interference suppression," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 2, pp. 190–193, Mar. 2012.

[35] R. Natsuaki, T. Motohka, M. Watanabe, M. Shimada, and S. Suzuki, "An autocorrelation-based radio frequency interference detection and removal method in azimuth-frequency domain for SAR image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5736–5751, Dec. 2017.

[36] T. Bollian, B. Osmanoglu, R. F. Rincon, S.-K. Lee, and T. E. Fatoyinbo, "Detection and geolocation of P-band radio frequency interference using EcoSAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3608–3616, Oct. 2018.

[37] N. Li, Z. Lv, and Z. Guo, "Pulse RFI mitigation in synthetic aperture radar data via a three-step approach: Location, notch, and recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[38] K. Dai *et al.*, "Monitoring activity at the Daguangbao mega-landslide (China) using Sentinel-1 TOPS time series interferometry," *Remote Sens. Environ.*, vol. 186, pp. 501–513, Dec. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.

[39] K. Yang, L. Yan, G. Huang, C. Chen, and Z. Wu, "Monitoring building deformation with InSAR: Experiments and validation," *Sensors*, vol. 16, no. 12, p. 2182, Dec. 2016.
[40] A. Chojka, P. Artiemjew, and J. Rapiński, "RFI artefacts detection in Sentinel-1 Level-1 SLC data based on image processing techniques," *Sensors*, vol. 20, no. 10, p. 2919, May 2020.
[41] P. Artiemjew, A. Chojka, and J. Rapiński, "Deep learning for RFI artifact recognition in Sentinel-1 data," *Remote Sens.*, vol. 13, no. 1, p. 7, Dec. 2020.
[42] X. Leng, K. Ji, and G. Kuang, "Radio frequency interference detection and localization in Sentinel-1 images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9270–9281, Nov. 2021.
[43] M. Tao, S. Lai, J. Li, J. Su, Y. Fan, and L. Wang, "Extraction and mitigation of radio frequency interference artifacts based on time-series Sentinel-1 SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.
[44] S-1 Mission Performance Center (S-1 MPC). (2022). *SAR-MPC Sentinel-1: Using the RFI Annotations*. [Online]. Available: https://sentinels.copernicus.eu/documents/247904/1653442/DI-MPC-OTH-0540-1-0-RFI-Tech-Note.pdf
[45] F. J. Meyer, J. B. Nicoll, and A. P. Doulgeris, "Correction and characterization of radio frequency interference signatures in L-band synthetic aperture radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 10, pp. 4961–4972, Apr. 2013.
[46] F. Bovolo and L. Bruzzone, "A detail-preserving scale-driven approach to change detection in multitemporal SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 12, pp. 2963–2972, Dec. 2005.
[47] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.
[48] Y. Zhang, Y. Dong, K. Wu, and T. Chen, "Hyperspectral anomaly detection with Otsu-based isolation forest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9079–9088, Sep. 2021.
[49] N. Li, Z. Lv, and Z. Guo, "SAR image interference suppression method by integrating change detection and subband spectral cancellation technology," *Syst. Eng. Electron.*, vol. 43, no. 9, pp. 2484–2492, Sep. 2021.
[50] X. Wang and C. Chen, "Ship detection for complex background SAR images based on a multiscale variance weighted image entropy method," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 2, pp. 184–187, Feb. 2017.
[51] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.
[52] L. Li, F. Zhang, Y. Shao, Q. Wei, Q. Huang, and Y. Jiao, "Airborne SAR radiometric calibration based on improved sliding window integral method," *Sensors*, vol. 22, no. 1, p. 320, Jan. 2022.
[53] M. Gong, S. Zhao, L. Jiao, D. Tian, and S. Wang, "A novel coarse-to-fine scheme for automatic image registration based on SIFT and mutual information," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4328–4338, Jul. 2014.
[54] P. Salembier, S. Liesegang, and C. Lopez-Martinez, "Ship detection in SAR images based on maxtree representation and graph signal processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2709–2724, May 2019.



Ning Li (Member, IEEE) received the B.S. degree in electronics information engineering from Northeast Forestry University, Harbin, China, in 2009, the M.S. degree in communication and information system from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2012, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences (IECAS), Beijing, China, in 2015.

In July 2015, he joined the Department of Space Microwave Remote Sensing System, IECAS, where he was an Assistant Professor. Since December 2017, he has been a Full Professor with the School of Computer and Information Engineering, Henan University, Kaifeng, China. His research interests include synthetic aperture radar (SAR) and inverse SAR imaging algorithms and autofocusing techniques, SAR polarimetric theory, and SAR image processing.

Dr. Li was a recipient of the Special Prize of President Scholarship for Postgraduate Students from the University of Chinese Academy of Sciences in 2015.



Hengrui Zhang was born in Henan, China, in 1990. He graduated from the School of Computer Network Technology, Zhengzhou University, Zhengzhou, China, in 2010. He is currently pursuing the M.S. degree with the School of Computer and Information Engineering, Henan University, Kaifeng, China.

His research interests include synthetic aperture radar (SAR) imaging, detection, and mitigation of radio frequency interference (RFI).



Zongsen Lv (Graduate Student Member, IEEE) was born in Henan, China, in 1996. He received the B.S. degree in computer science and technology from Henan University, Kaifeng, China, in 2014, where he is currently pursuing the M.S. degree with the School of Computer and Information Engineering.

His research interests include synthetic aperture radar (SAR) imaging and radio frequency interference (RFI) mitigation.



Lin Min was born in Henan, China, in 1963. He received the B.S. degree from the Huazhong University of Science and Technology, Wuhan, China, in 1984.

He then joined Henan University, Kaifeng, China, as an Assistant Professor, where he has been a Professor since 2013. The projects on SAR image applications and information processing system he participated in have been supported by the National Natural Science Foundation of China and the Department of Science and Technology of Henan Province. His research interests include SAR imaging algorithms and signal processing. He has published several papers, including peer reviewed journals, conference proceedings, patents, and software copyrights.



Zhengwei Guo (Member, IEEE) received the B.S. degree in radio technology engineering from the Huazhong University of Science and Technology, Wuhan, China, in 1984.

She then joined Henan University, Kaifeng, China, as an Assistant Professor, where she has been serving as a Professor with the School of Computer and Information Engineering since 2008. Her research focuses on management system software, data processing systems, and information security technology. Her projects on synthetic aperture radar (SAR) image applications and information processing system have been supported by the National Natural Science Foundation of China and the Department of Science and Technology of Henan Province. She has published several papers, including peer reviewed journals, conference proceedings, patents, and software copyrights.

Dr. Guo received the First-Prize Scientific and Technological Progress Award of Henan Province.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:17:30 UTC from IEEE Xplore. Restrictions apply.